CHARAN TEJA VALAVALA

SOFTWARE, DATA AND APPLIED AI

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GitHub | Portfolio | LinkedIn

PROFILE

BITS CSE undergraduate building software and applied AI projects in document search, financial data, and incident workflows. Also completed machine-learning and reinforcement-learning coursework and contributed to an education platform used by 200+ students and trainees.

SELECTED PROJECTS

Course Copilot — Python | FastAPI | MiniLM | FTS5 | Demo

- Built PDF/TXT document search with MiniLM embeddings, FTS5 BM25, 2:1 reciprocal-rank fusion, and page-level citations; optional Ollama answers check citation references.
- 2:1 hybrid hit 48/52 Hit@1 vs BM25-only 47/52 (MRR 0.962 vs 0.946); 14/15 out-of-scope prompts abstained on a 52-query, 13-note dev set.

FormScope — Python | FastAPI | SEC EDGAR | XBRL | Demo

- Built a FastAPI SEC explorer for XBRL facts, annual-period validation, fallback concepts, revenue growth, net margin, and source-linked 10-K passage search.
- Revenue and net income matched inline XBRL across 20 issuer filings: 40/40 fact checks.

IncidentPilot — Python | FastAPI | SQLite | Ollama | Demo

- Built a read-only incident-triage simulator with evidence-linked diagnoses, replayable SQLite traces, and approval-gated simulated actions.
- Added a 20-case synthetic Ollama evaluation harness for tool-call validity, evidence references, and fallback behavior.

COURSEWORK PROJECTS

Diabetic Retinopathy Detection — Python | OpenCV | scikit-learn

- Trained SVM classifiers on retinal fundus color and texture features with OpenCV; evaluated accuracy, precision, recall, and sensitivity.

Deep Reinforcement Learning for Object Tracking — Monte Carlo control | stochastic grid

- Built a stochastic grid with thermal states and teleportation; compared on-policy and off-policy Monte Carlo control.

EXPERIENCE AND EARLY-STAGE WORK

DataForge | Founder | concept stage — Feb-May 2026

- Explored an AI infrastructure and data center venture concept with mentor guidance; participated in early funding discussions.

MDDTI SBC | Web Development Intern — May-July 2024

- Contributed HTML/CSS frontend work to the organization website and online test platform, with MongoDB-backed features used by 200+ students and trainees.

EDUCATION AND ACHIEVEMENTS

BITS Pilani, Hyderabad Campus | B.E. Computer Science Engineering — Expected 2027+
AP EAMCET | Statewide rank 8
Narayana Junior College | Class XII | 98.1% — 2022
Narayana High School | Class X | 10/10 GPA — 2020

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript, SQL, HTML, CSS
Data and ML: NumPy, Pandas, Matplotlib, scikit-learn, OpenCV, Sentence Transformers, BM25/RRF
Backend and data: FastAPI, REST APIs, MongoDB, SQLite, SEC EDGAR/XBRL
Tools: Git, Docker, Ollama

RELEVANT COURSEWORK AND LANGUAGES

Artificial Intelligence, Machine Learning, Reinforcement Learning, Foundations of Data Science, Data Structures and Algorithms, Database Systems, Object-Oriented Programming. Spoken: English, Hindi, Telugu.